

Stream Habitat Fragmentation Caused by Road Networks in Spanish Low-order Forest Catchments

Jorge García Molinos

School of Natural Sciences, Department of Zoology
Trinity College, Dublin 2, Ireland

1. Introduction

Designing and maintaining optimal road networks is one of the most crucial factors in forest management influencing cost-effective timber harvesting, fire control and safety, forest inventory and monitoring, game control, residential access and recreational use in mountain watersheds (Lugo and Gucinski, 2000; Janowsky and Becker, 2003; Demir et al., 2009). Planning for multi-functional road networks with minimum environmental impacts is an essential aim of forest management nowadays (Gumus et al., 2008; Eastaugh and Molina, 2011). However, forest roads continue to be a major source of impact on natural ecosystems (Spelleberg, 1998; Trombulak and Frissell, 2000; Seiler, 2001). In the United States, around one fifth of its surface area is ecologically impacted by roads (Forman and Alexander, 1998). Forest managers need an improved understanding about the potential environmental impact of road networks in different settings on which to base appropriate planning, maintenance and decommissioning of roads. In this sense, studying the impact of roads on stream ecosystems is a good compromise solution. Streams are especially suitable indicators of environmental change because they are extremely sensitive systems that integrate changes occurring over the entire catchment area (Williamson et al., 2008).

Roads affect stream ecosystems in a variety of ways including alteration of hydrologic regimes (King and Tennyson, 1984; LaMarche and Lettenmaier, 2001) and increased sediment supply to streams (Luce and Black, 1999; Jones et al., 2000). Habitat fragmentation caused by road-stream crossings is a

subject of major concern and a priority in restoration and conservation management (Zwick, 1992; Roni et al., 2002; Gibson et al., 2005). Roads and streams share many characteristics; both compose long, linear networks conveying material and organisms across landscapes. This constrained linear nature increases their susceptibility to blockage, while increasing the frequency of intersections. Further, the steep topography of mountain watersheds often forces road delineation to occur near water courses, thereby increasing the number of crossings (Roni et al., 2002).

Stream barriers commonly occur when culverts are used instead of bridges (Warren and Pardew, 1998; Benton et al., 2008). This is common in forest roads because culverts are more cost-effective (Gibson et al., 2005). Hydraulic efficiency and road protection guide normal culvert design criteria (Norman et al., 2001). However, culverts based purely on engineering criteria can severely impair stream ecosystems by blocking the passage of aquatic organisms and altering or limiting important stream processes such as the movement of sediment, woody debris, organic carbon and nutrients (Bates et al., 2003; Bates et al., 2008). Poorly designed culverts are typically narrower than the width of the natural channel, have a skewed alignment with the stream, are improperly graded or present steep slopes and smooth interior surfaces to increase drainage capacity (Fig. 1). Undersized and poorly aligned



Fig. 1. Examples of common culvert barriers (photos J. García Molinos): (A) large volumes of sediment and debris accumulated upstream of an undersized culvert, (B) poorly graded corrugated pipe culvert creating a depth barrier, and (C) multi-box culvert with perched outlet.

culverts are often responsible for abrupt stream-to-culvert transitions. Depending on flow conditions, this can create excessive flow velocities and turbulence at the culvert entrance or obstruct the flow producing a backwater effect that favours the deposition of sediment and woody debris. Poorly graded culverts, steep slopes and smooth surfaces create problems related to the hydraulic conditions (velocity and depth) inside the culvert barrel. A very common related issue is downstream streambed scouring resulting in culverts with perched outlets.

Studies carried out in North America demonstrate that the barrier problem associated with culvert crossings is ubiquitous and dramatic (Table 1). Millions of culvert crossings block the passage of resident and anadromous fish to highly valuable headwater stream habitats in mountain watersheds. Beechie et al. (1994) found, for example, that 13% of historical salmon habitat was blocked by culverts in a large river catchment in Washington. This was estimated to account for a 30-58% reduction in smolt production. This wealth of information is lacking in European countries where, though awareness of the problem exists (Glen, 2002; Larinier, 2002; García Molinos et al., 2004), its extent and implications remain largely unknown.

Here, the author reports on the results of a field study carried out to (i) investigate the number and type of road-stream crossings, in relation to both road and stream typology, in small low-order forest catchments in northern Spain, (ii) identify the proportion of culvert barriers along roads and describe their causes, (iii) quantify the amount of stream habitat above culverts that is fragmented from the principal water course, and (iv) outline possible best management practices and mitigation measures in the context of current advances in the field.

Table 1: Literature citations on the proportion of culverts limiting or blocking the passage of fish (salmonids) for different North American studies

<i>Reference</i>	<i>Location</i>	<i>Number of culverts</i>	<i>Percentage of barriers</i>
(Taylor, 2001)	California	28	46
(Chesnut, 2002)	British Columbia	31	85-90
(Taylor et al., 2002)	California	51	90-98
(Tchir et al., 2004)	Alberta	142	61-74
(Gibson et al., 2005)	Labrador	47	53
(Lang, 2005)	California	312	92
(Cahoon et al., 2005)	Montana	47	81
(Park et al., 2008)	Alberta	374	50*
(Poplar-Jeffers et al., 2009)	Virginia	120	97
(Price et al., 2010)	Washington	77	30

* Only perched culverts surveyed as barriers.

2. Methods

2.1 Study Area

This study was conducted over an area draining a small forest sub-catchment (6842 ha) comprising the upper-middle part of the Arga River (upstream/downstream co-ordinates ED50 42.936585N, -1.502509W; 42.851971N, -1.583426W) in the Spanish region of Navarra (Fig. 2). The river system is composed of the main stem, the Arga River, and a tributary network comprising several high-gradient (2-16.7%) small first-to-second-order streams. The ecological status of the Arga River within the study area has been recently classified as “good” following the criteria set by the Water Framework Directive (Castiella Muruzábal et al., 2007). These water courses are classified as mixed salmonid waters hosting cool water (mean annual temperature < 15 °C) fish communities, dominated by brown trout (*Salmo trutta*), stone loach (*Noemacheilus barbatulus*) and minnows (*Phoxinus phoxinus*). Land use is represented predominantly by pine (*Pinus sylvestris*) reforestations alternated with natural oak woods and pasture land. Urban development in the area is restricted to several small villages scattered along the valley of the Arga River. However, human activity is high due to the proximity of the city of Pamplona.

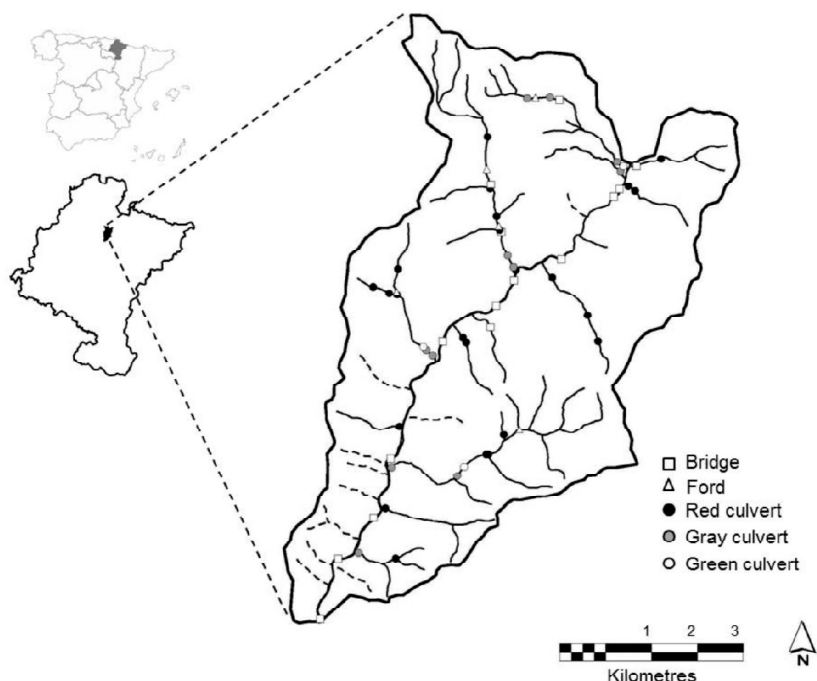


Fig. 2. Map of study area showing crossing locations by barrier status. Dotted lines indicate dried stream channels that were not included in the survey.

2.2 Survey Design and Barrier Analysis

Road-stream intersections within the study area were initially located on a 1:25,000-scale georeferenced topographic map using geographic information system tools (gvSIG 1.11.0). The georeferenced cartographic material was provided by the Geographic Information National Centre, Spanish Ministry of Infrastructures (Centro Nacional de Información Geográfica, Ministerio de Fomento), which can be downloaded at <http://centrodedescargas.cnig.es/CentroDescargas/index.jsp>. All crossings were visited during April 2010 under normal flow conditions. Upon locating sites in the field, crossings were classified according to their typology and grouped by stream order and road category. Roads were divided into three categories depending on the costs associated with road construction: class I included paved roads with or without associated road-side drainage elements; class II grouped unpaved roads with compacted platforms with or without drainage elements; and class III comprised unpaved minor roads with uncompacted platforms and no drainage elements. Crossings located on dried channels or on non-water courses (e.g., ditches or cross drains) were excluded at this stage.

Physical information was collected at each culvert crossing to be used later for barrier classification. Most data were obtained from a longitudinal survey of the culvert and the adjacent stream channel (Fig. 3) and included the collection of data at several specific points with a surveying optical automatic level (Nikon AX-2S, Japan). In addition to the longitudinal survey, other culvert-specific information included barrel size (diameter or width), construction material and typology (e.g., pipe, open arch, box). Finally, the average active channel width ($n = 5$) was measured upstream away from the zone of influence of the culvert.

To analyse the barrier status of each culvert crossing the author used the fish passage ranking protocol developed by Taylor and Love (2003). This protocol was developed to assess the passage status for resident salmonids at

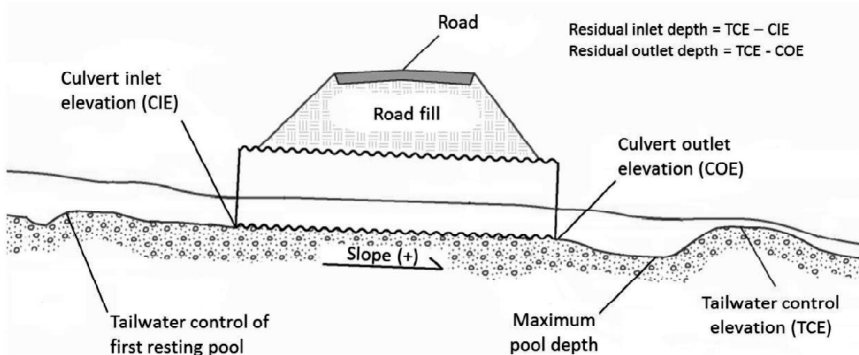


Fig. 3. Schematic longitudinal view of a culvert crossing showing the principal surveyed points used in the barrier classification criteria.

Table 2: Classification criteria to identify the barrier status of culvert crossings, as determined by resident salmonid passage, in the study

<i>Category</i>	<i>Status</i>	<i>Criteria</i>
Green	No barrier. Passage of all salmonid species and life stages.	Streambed substrate throughout culvert AND Inlet width $\geq$ active channel width OR no outlet drop AND residual inlet depth $\geq$ 0.15 m
Gray	Potential barrier. Passage likely to be impeded. In-depth analysis required.	All other intermediate cases
Red	Barrier. Culvert fails to meet minimum passage criteria.	Outlet drop $\geq$ 0.61 m OR barrel slope $>$ 3% AND contains no waffles AND streambed substrate throughout culvert absent

After Taylor and Love 2003.

culvert crossings and it compares culvert-specific physical characteristics with fish passage criteria, classifying culverts into one of three passage type categories (Table 2). Residual inlet and outlet depths are used respectively as descriptors of the minimum water depth throughout the culvert and the existence of perched outlets (negative values) at very low flow conditions (Fig. 3). In more exhaustive studies (Taylor, 2001; Lang, 2005), this method is used normally as a first classification filter to reduce the number of crossings (Gray culverts) which need to be assessed through more sophisticated and time-consuming analyses. However, it is considered an appropriate approach for the purposes of this study because it gives an easy-to-interpret classification by placing culvert crossings along a barrier gradient (green to red). It is important to note that the Red category does not necessarily define culverts blocking passage in an absolute manner, but rather culverts failing to meet minimum fish passage criteria requirements as outlined by Californian fish regulations. Passage criteria for this protocol are selected to accommodate passage of the weaker swimming individuals within each species and life stage of resident and anadromous salmonids in accordance with these regulations (Taylor and Love, 2003).

2.3 Habitat Fragmentation Assessment

The proportion of fragmented habitat upstream of culvert barriers (Gray and Red culverts) was calculated through GIS analysis by dividing the total stream length from the barriers to their sources by the total length of streams surveyed. In those cases, where more than one culvert crossing blocked access to the same stream, the most downstream barrier was selected for calculations.

3. Results

3.1 Crossing Density and Typology

A total of 69 stream crossings were initially identified on the map. Subsequent site visits reduced this number to 57, as 12 crossings occurred on dried channels or on non-water courses and were not considered in this survey. Culverts (37/57 or one culvert every 1.97 km of stream) were by far the most abundant crossing structures (Fig. 4A). The majority were concrete (98%), circular (62%) culverts. Other frequent types included box (22%) and open arc (11%) culverts. Bridges (26%) and stream fords (9%) were other crossing structures present in the study area. However, proportions varied considerably when categorising crossings by road typology (Fig. 4A). Concentrated over the Arga River, bridges were most frequently used in paved roads (class I), accounting for 73% of the total numbers of these type of crossings. At the other extreme, stream fords occurred invariably in unpaved minor forest roads (class III).

3.2 Barrier Status

Only 8% (3/37) of the surveyed culverts were found to meet the minimum criteria to guarantee resident salmonid fish passage for all species and life-stages (Green). Among the rest, 30% crossings were classified as potential (Gray) barriers and 62% as complete (Red) barriers. A continuous layer of natural stream bed substrate throughout the culvert barrel was only found in three culverts, representing the passage criterion most frequently violated (92%; Fig. 4B), followed by undersized culverts (89%), excessive slope (60%), insufficient residual inlet depth (57%), and existence of perched outlets (51%; 30% over 0.6 m). Further, 97% of the culverts failed to meet two or more criteria. Barrier status differed significantly between stream orders (Mann-Whitney $U = 72$, $n_1 = 16$, $n_2 = 21$, $p = 0.001$). Second-order streams (Fig. 4D) accumulated all Green crossings and had much larger Gray-to-Red barrier ratios as compared to first-order streams. No significant differences ($p > 0.05$) in barrier proportions were found between road classes (Fig. 4C), though paved roads (class I) had higher proportions of Green culverts and lower proportions of Gray and Red culverts than unpaved minor forest roads (class III).

The accumulated frequency curves of the different culvert physical characteristics followed distinctive and contrasting patterns (Fig. 5). The accumulated curve of the inlet-active channel width ratio (0.31-1.14), which should be ≥ 1 to avoid constriction of the natural channel, presented a sigmoid pattern with a high frequency of values around 0.57-0.67. On the contrary, the culvert slope (1.1-14.3%; Fig. 5B) followed an exponential curve tailing off at values above the slope limit (3%). Finally, the outlet drop (0-1.18 m) curve displayed a more linear pattern with several small frequency breaks (Fig. 5C).

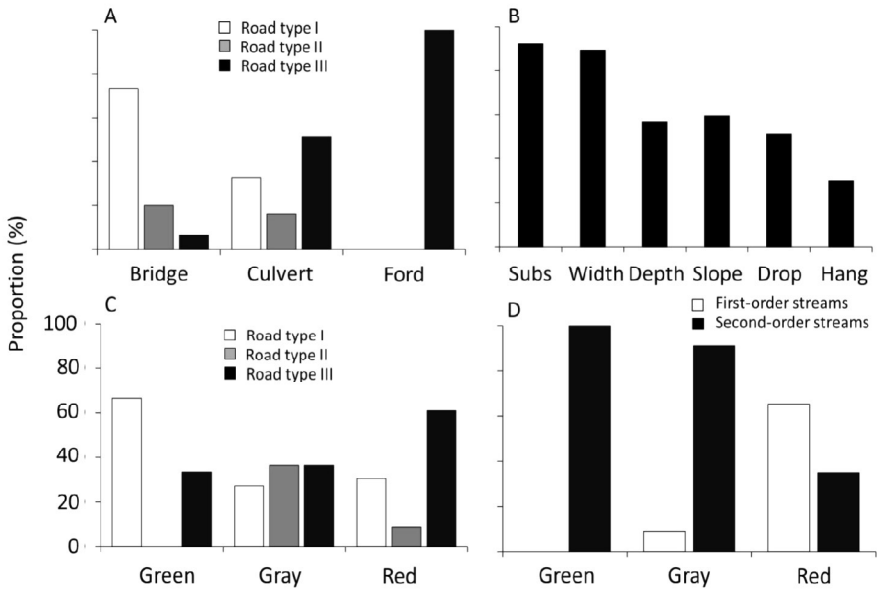


Fig. 4. Proportions of (A) road crossings grouped by crossing type and road class, (B) culverts failing to meet different fish passage criteria (Subs: no continuous streambed substrate; Width: culvert inlet width < active channel width; Depth: residual inlet depth < 0.15 m; Slope: barrel slope > 3%; Drop: existence of outlet drop; Hang: drop $\geq$ 0.61 m), (C) culverts grouped by barrier status and road class, and (D) culverts grouped by barrier status and stream order.

3.3 Habitat Fragmentation

The study area comprised 72.8 km of stream, of which 13.8 km corresponded to the Arga River and 59 km to the tributary network comprising first- and second-order streams. Of these, 53.8 km (74% of the total stream length or 91% of the tributary network) were located upstream of barrier (Gray or Red) crossings, isolated from the main stem. Because the main road in the study area (N-135) runs across the valley, parallel to the Arga River, almost all tributaries were affected by a crossing near their mouths (Fig. 2). Of these (13), two were bridges, one was classified as an open crossing (Green) and five as potential (Gray) culvert barriers, while the remaining five were categorized as complete (Red) barriers. However, fragmented stream habitat was characterised by a high degree of overlap among culvert barriers resulting in only a 9% of the tributary network accessible from the main stem (Fig. 2). Despite the high degree of fragmentation overlap among culvert barriers, some culverts were responsible for a significant proportion of the fragmented habitat. Seven culverts (19%) accounted for 29% of the fragmented stream habitat.

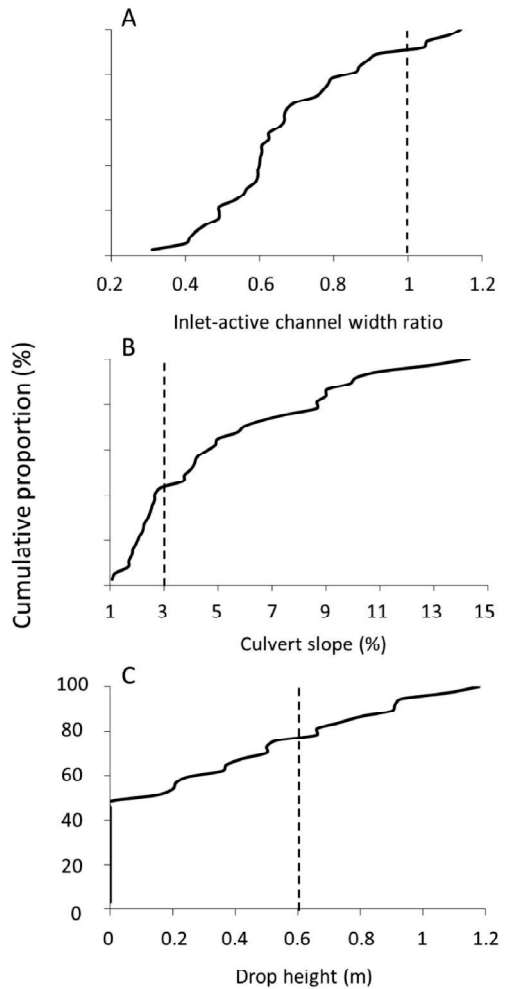


Fig. 5. Accumulated frequency distribution of (a) culvert inlet-channel width ratios, (b) outlet drop, and (c) culvert slope. Broken vertical bars indicate maximum limits set by fish passage criteria.

4. Discussion

This study represents the first published account of the extent, typology and potential consequences of culvert barriers associated with forest road networks in Spain. Culvert crossings within the study area were associated predominantly with low-order streams and low-category roads reflecting the advantages provided by these structures in relation to cost and ease of installation as compared to bridges. Surprisingly, culvert density in the study area (one culvert every 1.97 linear km of stream) appeared to be much higher than that reported in other studies (e.g., Tchir et al., 2004; Poplar-Jeffers et al., 2009).

However, this could be a consequence of our surveyed stream network, mainly comprising high-gradient (70% of the culverts placed in stream reaches above 3% of slope) first- and second-order streams. The extremely high proportion of potential culvert barriers is clearly symptomatic of poor environmental design and lax installation practices. Poorly graded, undersized, steep culvert barrels were commonplace, as appears to be the case elsewhere (e.g., Gibson et al., 2005; Park et al., 2008; Poplar-Jeffers et al., 2009). Excessive slope was, however, the barrier criteria associated to Red crossings most frequently violated within the study area, well above (+30%) excessive (>0.6 m) drop height. This situation is common in steep gradients when culverts are installed following the natural stream gradient (Poplar-Jeffers et al., 2009). Further, whereas the significant differences in culvert barrier status found between first- and second-order streams may also reflect certain degree of design relaxation associated to stream size, the absence of significant differences between road categories apparently indicate no road budgeting constraints associated to barrier generation.

The profusion of culvert crossings and the associated, disproportionately high, frequency of barriers made the proportion of tributary stream habitat potentially isolated from the main stem watercourse to be strikingly high (91%). Though the tributary network within the study area is composed exclusively of small low-order streams, this situation is likely to hinder significantly aquatic communities within the Arga River watershed. Streams are extremely sensitive to habitat fragmentation because they are highly dynamic ecosystems with strong longitudinal hierarchical dependence with regard to movements of aquatic biota and materials (Vannote et al., 1980).

Small headwater streams play an often overlooked, but highly important, multi-scale role within the watershed (Gomi et al., 2002). Forested headwater streams receive a predominant input of allochthonous organic material (e.g., leaf litter and woody debris) that is often accumulated and retained in headwater channels for long periods of time, allowing for the gradual decomposition of coarse particulate matter deposits to smaller particles that are then exported downstream to higher order streams (Wallace et al., 1991; Kiffney et al., 2000). A productive and distinctive macroinvertebrate community inhabits these banks of organic matter (Wallace et al., 1997), representing an important source of food for the salmonids that often spawn and nurse in these streams (Erkinaro and Gibson, 1997; Klemetsen et al., 2003). Isolation of these stream habitats from foraging grounds in larger rivers and lakes can lead to strong reductions in fish production (Beechie et al., 1994; Petty et al., 2005).

Despite the alarming nature of the figures, barrier estimation in this study might well be conservative. The protocol used to assess culvert barrier status was developed for the passage of resident salmonids in Californian streams, based on current fish passage regulations (Taylor and Love, 2003). State, regional and local regulations directing the design of water-crossing structures for roads normally include provision for fish passage (Moore et al., 1999).

However, enforced passage criteria commonly focus on salmonid species because of the migratory behaviour characteristic of these species and their economic relevance. Passage conditions for other fish species with weaker swimming and leaping capacities are thus often neglected. Further, in principle, habitat fragmentation can have deleterious effects on any aquatic species (Zwick, 1992; Vaughan, 2002) or even terrestrial fauna using stream corridors (Yanes et al., 1995). Movement is an essential mechanism to most aquatic organisms (Jackson, 2003). Animals disperse over different ranges and at varying times during their lives to fulfill life-history and metabolic needs such as reproduction, nursing, growth, feeding, over-wintering, dispersal and colonization. Movements are also important avoidance mechanisms in response to changes in habitat conditions caused by natural or anthropogenic disturbances (Schlosser and Angermeier, 1995). Preserving the connectivity of a stream network is also crucial for maintaining genetically diverse populations (Neraas and Spruell, 2001). Fragmented populations are more vulnerable to inbreeding and genetic drift, which decrease population fitness and evolutionary adaptability (Lande, 1988; Woodruff, 1990). Given the narrow, linear nature of stream ecosystems, unrestricted movement is critical for maintaining viable populations (Morita and Yokota, 2002).

Management approaches to allow unrestricted movement should start from avoiding construction or removal of crossings. Where permanent crossings are necessary, culverts may be replaced by less-problematic structures such as bridges. Culvert barrier replacement has been shown to be a very effective measure. Recolonization and spawning in isolated reaches usually occur shortly after replacement (Shrimpton et al., 2008), though the magnitude of the response is intimately linked with the quality of the restored habitat (Pess et al., 2005). If a culvert is to be constructed, stream simulation currently directs adequate design criteria for environment-friendly culvert crossings and should be the first choice (Bates et al., 2003; Bates et al., 2008). This relatively new methodology is based on ecosystem-based criteria seeking to mimic the morphological characteristics of the stream channel inside the culvert to guarantee free and unrestricted passage to all aquatic species and to maintain fundamental ecosystem functions such as sediment and debris transport. The diversity and complexity of the natural channel is replicated by matching the culvert to the full channel width and creating a continuous and stable substrate layer throughout the culvert, following the morphology and gradient of the natural channel. In this way, constriction of the natural channel is avoided, while retaining flow depths and velocities inside the culvert similar to those encountered in the stream.

Dispersal barriers are a major cause of fish population decline (Jager et al., 2001; Roni et al., 2002; Gibson et al., 2005), altered ecosystem functioning (Layman et al., 2007) and represent a global threat to biodiversity (Zwick, 1992; Dynesius and Nilsson, 1994). Though regulations enforcing aquatic passage criteria for water-crossing structures are uncommon in Europe, there

is a growing interest on the subject amongst managers and policy-makers. A principal motivation behind this interest is the importance of preserving and restoring stream connectivity as a necessary step to achieve the 2015 “Good Ecological Status/Potential” objective for surface water bodies required by all EU Member States through the European Union Water Framework Directive (Directive 2000/60/EC). Numerous assessment protocols and guidance documents for fish passage at road-stream crossings have appeared over the last decade in Europe (Ministerio de Medio Ambiente, 2006; Kemp et al., 2008; Ministerio de Medio Ambiente y Medio Rural y Marino, 2008; Balkham et al., 2010). However, experience has shown the limitations of similar documents that were previously enforced to prevent this situation (Price et al., 2010).

A holistic, multidisciplinary approach involves improved ecosystem-based design, construction compliance, and long-term management planning. Management plans need to be founded on an exhaustive inventory to assess the condition of existing crossings and analyse their environmental implications at the catchment-scale. This information will allow the establishment of prioritization protocols directing the decommissioning and correction of barriers and the implementation of appropriate maintenance practices. Detailed information on existing crossings and sensitive areas will also aid to conceive a long-term road network that minimises environmental impacts and associated costs. Stream restoration and conservation require active incorporation of the multi-scale continuum concept into management (Fausch et al., 2002). Reconnecting fragmented habitat means restoring stream ecological function and should, consequently, be a primary management objective (Bradshaw, 1996; Gibson et al., 2005).

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